
Conformal Source Regions for Sparse-Sensor Pollution Localization: A Benchmark with Exact Posteriors that Audits — and Trains — Learned Localizers

Anonymous Author(s)

Affiliation

Address

email

Abstract

When a gas leak or pollution release happens, responders must answer a deceptively simple question: *where is it coming from?* — increasingly from only a handful of cheap, fixed sensors. A learned localizer can output a heatmap of likely sources, but a heatmap alone does not say how much to trust it. We introduce a benchmark for uncertainty-aware source localization whose defining asset is the *exact Bayesian posterior* of every scenario, shipped with the data, and show that this asset does double duty. **First, it audits.** Pairing split conformal prediction with the exact posteriors yields a preregistered three-way test of any localizer: calibration (do 90% regions cover 90%?), sharpness (are regions as tight as the data physically permit?), and a causal information dose–response (do regions track added sensors, geometry, and noise the way the physics dictates?). On a standard DeepSets baseline the verdict is differentiated: calibrated (conformal-on-exact-posterior attains 90.7%; the learned localizer 89.6%, with 87.5% for a second seed quantified as a finite-sample fluctuation), but a median $5.6\times$ too wide — the network, not the data, is the bottleneck — and directionally correct on every knob at only half the oracle’s magnitude. **Second, it trains.** Using the same shipped posteriors as a *physics-derived training signal* — distilling each scenario’s exact posterior, mildly tempered, into the identical architecture on identical data — sharpens regions to $3.9\times$ and nearly restores oracle-proportional sensor response ($0.47 \rightarrow 0.87$ slope ratio); raw (untempered) physics targets fail, an optimization finding in its own right. The study also surfaced a floating-point failure mode that silently destroys conformal coverage on sharp probability maps, among other portable implementation lessons. To our knowledge this is the first benchmark in this domain to ship exact per-scenario posteriors, the first to pair finite-sample conformal regions with an exact-posterior sharpness audit, and the first to test learned localization uncertainty causally through a controlled dose–response protocol.

1 Introduction

Locating the source of an airborne pollutant from a sparse sensor network is a classic inverse problem whose legal stakes are rising. Under the U.S. EPA Super-Emitter Program, certified third parties notify operators of remotely detected methane events (≥ 100 kg/hr), and an operator within 50 m of the reported coordinates must investigate and report [U.S. EPA, 2024]; EU Regulation 2024/1787 mandates measurement, reporting, and verification of energy-sector methane [European Union, 2024]; and UNEP’s Methane Alert and Response System delivered 1,200 satellite-based notifications in its first two years — of which one per cent were responded to [UNEP, 2024]. When attributing an emission to a *location* carries consequences, the *uncertainty* of that attribution must itself be

trustworthy: a localizer that is confidently wrong is worse than one that honestly reports “somewhere in this larger region.”

Three properties matter. **Calibration** (trustworthiness): a claimed 90% region should contain the true source 90% of the time. **Sharpness** (actionability): among calibrated regions, smaller is better, and a region should be as tight as the data physically allow — a large region is itself a signal that the sensor network is inadequate. **Dose–response stability** (trust under change): sensors are added and lost constantly, so a region should shrink when information arrives and grow when it is removed, tracking what an oracle would do.

Contributions. Three, in order of emphasis: **(1) A benchmark whose central asset is exact posteriors.** A controlled advection–diffusion simulator with an analytic Green’s-function forward model, a frozen 14,000-scenario benchmark, constructed dose–response sets — and, for every scenario, the *exact grid posterior* of the source location, shipped with the artifact. The posteriors power a preregistered three-way audit of any localizer: finite-sample conformal calibration, a sharpness audit against the physical information limit, and a causal information dose–response protocol. **(2) A preregistered audit of a standard recipe.** We deliberately assemble off-the-shelf components — DeepSets [Zaheer et al., 2017] and split conformal prediction [Angelopoulos and Bates, 2021] — and measure whether the resulting uncertainty is calibrated, sharp, and stable (it is calibrated; it is not sharp; Sec. 6.1–6.3). **(3) The posteriors double as a training signal.** Distilling each scenario’s exact posterior — tempered to a learnable sharpness — into the *identical* architecture on *identical* data substantially improves every uncertainty-quality metric the audit measures (exploratory, validation-selected; Secs. 4.5, 6.4). Verbatim: *To our knowledge, this is the first public benchmark in this domain to ship exact per-scenario posteriors, the first study to combine finite-sample conformal spatial regions with an exact-posterior sharpness audit for sparse-sensor atmospheric advection–diffusion source localization, and the first to test learned localization uncertainty causally via a controlled information dose–response protocol against exact posteriors.* We claim no new conformal algorithm, no new architecture, no coverage under distribution shift, no field deployability, and no abstention capability; the distillation recipe is an application of teacher–student learning whose teacher is exact physics rather than another network.

Findings at a glance. (i) **H1:** the conformal machinery is sound — applied to the exact posterior it attains 90.7% coverage; the learned localizer attains 89.6% (seed 2) and 87.5% (seed 1, a quantified $p = 0.0045$ finite-sample fluctuation; Sec. 6.1). (ii) **H2:** calibrated is not sharp — learned regions are a median $5.6\times$ larger than the data permit; the network, not the data, is the bottleneck (Sec. 6.2). (iii) **H3:** the learned regions respond in the physically correct direction on all three information knobs, at roughly half the oracle’s magnitude (Sec. 6.3). (iv) **The benchmark trains as well as it audits:** distilling its shipped exact posteriors — mildly tempered — into the *same* model on the *same* data sharpens regions to $3.9\times$ and lifts the sensor-response slope ratio from 0.47 to 0.87; raw physics targets fail (Secs. 4.5, 6.4). (v) Three implementation lessons that transfer beyond this benchmark: the standard conformal score silently saturates in floating point on sharp heatmaps; fixed-node rate quadrature cannot marginalize peaked likelihoods; and at this data scale the specified architecture memorizes unless the problem’s exact symmetry group is injected as augmentation (Secs. 4.2–4.3, 4.1).

2 Related Work

Source term estimation and Bayesian inversion. Estimating the location and rate of an atmospheric release is reviewed by Hutchinson et al. [2017]. Bayesian/MCMC plume inversion [Newman et al., 2024] quantifies uncertainty rigorously but is per-scenario and heavy, and its credibility depends on the assumed dispersion model. Physics-informed neural networks recover advection–diffusion sources [Chuprov et al., 2025, Anague et al., 2025] but provide no finite-sample coverage. “Information density” in inverse problems [Bangerth et al., 2022] formalizes why some regions are recoverable and others are not.

Conformal prediction for localization and inverse problems. Conformal prediction [Angelopoulos and Bates, 2021] has recently reached localization: acoustic source localization with Gaussian processes [Rozenfeld and Laufer-Goldshtein, 2025]; task-driven UQ for imaging inverse problems

Table 1: Positioning matrix. $\checkmark$ = addressed; $(\checkmark)$ = partially. Axes: A amortized inference, B finite-sample region coverage, C audit vs. exact/reference posterior, D controlled causal test of uncertainty, E sparse atmospheric sensors.

Work	A	B	C	D	E
PINN plume inversion [Chuprov et al., 2025, Anague et al., 2025]	$(\checkmark)$				$\checkmark$
Bayesian/MCMC STE [Newman et al., 2024]			$(\checkmark)$		$\checkmark$
Conformal acoustic loc. [Rozenfeld and Laufer-Goldshtein, 2025]	$(\checkmark)$	$\checkmark$			
Conformal imaging UQ [Wen et al., 2024]	$(\checkmark)$	$\checkmark$			
Conformal network detection [Jian et al., 2026]	$(\checkmark)$	$\checkmark$			
Distill-Belief [Shi et al., 2026]	$\checkmark$		$\checkmark$		$(\checkmark)$
CP4SBI [Cabezas et al., 2025]	$\checkmark$	$\checkmark$			
This work	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$

[Wen et al., 2024]; multi-source detection on networks [Jian et al., 2026]; anatomical landmark localization [Jonkers et al., 2025]. Edwards and Smith [2025] treat radiological plume identification with categorical and Bayesian networks. Distill-Belief [Shi et al., 2026] distills a Bayes-correct particle-filter teacher for closed-loop source localization — closest in spirit to our post-hoc physics-distilled variant (Sec. 6.4). Closest in machinery, CP4SBI [Cabezas et al., 2025] conformally calibrates simulation-based-inference credible sets, but with no atmospheric domain, no exact-posterior sharpness audit, and no dose–response protocol.

Positioning. Along five axes — (A) amortized inference, (B) finite-sample region coverage, (C) audit vs. an exact/reference posterior, (D) controlled causal test of uncertainty, (E) sparse atmospheric sensors — prior work occupies subsets (Table 1); this study is the first to occupy all five simultaneously.

3 Problem Setup and Physics

A scalar concentration $C(\mathbf{x}, t)$ on $\Omega = [0, 1]^2$, $t \in (0, 1]$, is transported by a constant wind $\mathbf{u} = (u, v)$ and isotropic diffusion D :

$$\frac{\partial C}{\partial t} + u \frac{\partial C}{\partial x} + v \frac{\partial C}{\partial y} = D \left(\frac{\partial^2 C}{\partial x^2} + \frac{\partial^2 C}{\partial y^2} \right) + S(\mathbf{x}, t), \quad \mathbf{x} \in \Omega, t \in (0, 1]. \quad (1)$$

For a continuous point release at $\mathbf{x}_s$ switched on at $t = 0$, regularized by a small source blob of width $\sigma_s = 1/128$, the unit-rate response is the superposition of spreading, advected Gaussians:

$$A(\mathbf{x}, t; \mathbf{x}_s, \mathbf{u}, D) = \int_0^t \frac{1}{\pi w_s} \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}_s - \mathbf{u} s\|^2}{w_s}\right) ds, \quad w_s = 4Ds + 2\sigma_s^2. \quad (2)$$

The age integral is evaluated by 64-node Gauss–Legendre quadrature after the log-age substitution $s = t e^{-\tau}$, $\tau \in [0, 12]$ (bound tuned empirically; Appendix A); responses are generated in `float32` and validated in `float64`. A release of rate q produces field qA .

Observation model. Sensor i at $\mathbf{x}_i$ takes 30 readings at $t_k = k/30$:

$$y_{ik} = \frac{q A(\mathbf{x}_i, t_k; \mathbf{x}_s, \mathbf{u}, D)}{C_{\text{ref}}} + \varepsilon_{ik}, \quad \varepsilon_{ik} \sim \mathcal{N}(0, \sigma^2), \quad (3)$$

with independent Bernoulli reading dropout; dropped readings are *omitted*, never imputed, so the model must handle variable-length inputs. The data-generation prior is in Table 2. Splits are by *complete scenario* to prevent leakage (Sec. 5).

4 Method

4.1 Amortized posterior (DeepSets classifier)

The localizer is a permutation-invariant DeepSets network [Zaheer et al., 2017] of 4,869,568 parameters, chosen deliberately as a standard, unremarkable architecture. Each retained reading is tokenized

Table 2: Benchmark data-generation prior. All quantities sampled independently per scenario. $\text{LogUniform}(a, b)$ is uniform in log-space.

Quantity	Symbol	Distribution
Source location	$\mathbf{x}_s$	$\text{Uniform}([0.1, 0.9]^2)$
Release rate	q	$\text{LogUniform}(0.5, 5)$
Wind speed	$\ \mathbf{u}\ $	$\text{Uniform}(0.5, 2.0)$
Wind direction	θ_u	$\text{Uniform}(0, 2\pi)$
Diffusivity	D	$\text{LogUniform}(0.002, 0.02)$
Sensor count	N	Uniform integer $\{4, \dots, 12\}$
Sensor positions	$\mathbf{x}_i$	i.i.d. $\text{Uniform}([0, 1]^2)$
Noise std	σ	$\text{LogUniform}(0.01, 0.10)$
Reading dropout	p_{drop}	$\text{Uniform}(0, 0.20)$
Observation times	t_k	$k/30, k = 1, \dots, 30$
Train / Val / Calib / Test		10,000 / 1,000 / 2,000 / 2,000
Dose-response base scenarios		50 (12-sensor master layouts)

as $\mathbf{z}_{ik} = [\text{Fourier}(x_i), \text{Fourier}(y_i), \text{Fourier}(t_k), \text{asinh}(y_{ik}/\sigma)]$ with frequencies $\{1, 2, 4, 8\}$. A per-token MLP ($\rightarrow 256 \rightarrow 256$, GELU + LayerNorm) embeds each token; embeddings are pooled by *both* masked mean and masked max, concatenated; a context MLP ($5 \rightarrow 64 \rightarrow 64$) encodes $(u, v, \log D, \log \sigma, N)$; a decoder ($576 \rightarrow 1024 \rightarrow 4096$) produces a softmax $P_j(\cdot)$ over a 64×64 grid. Training minimizes cross-entropy to a smoothed target (Gaussian bump, std 0.75 cells, truncated at 3σ , centered on the true source): AdamW, lr 3×10^{-4} cosine-decayed to 3×10^{-5} , weight decay 10^{-4} , batch 256, ≤ 200 epochs, early stopping on validation NLL (patience 25), mixed precision, two seeds. Because targets are sampled from the prior, the softmax is an *amortized approximation to the grid posterior* $p(c | \mathbf{y})$.

Amendment: symmetry augmentation (deviation). As preregistered, this recipe *memorizes*: training NLL falls to 1.5 while validation NLL never beats the uniform level (≈ 7.96 ; $\log 4096 = 8.32$) and validation MAP error (0.44) is worse than the peak-sensor baseline (0.37) — verified not attributable to mixed precision, learning rate, or the context branch. We therefore apply to each batch a random D4 symmetry of the unit square (rotation $k \cdot 90^\circ$ + reflection) jointly to sensors, wind, and source — an *exact* physics symmetry (validated to 2×10^{-13}) under which the prior is invariant, so augmented scenarios are exact prior draws. Nothing else changes; validation NLL then tracks training loss with no overfitting gap. Finding: at this data scale the benchmark *requires* symmetry augmentation to be learnable by the specified architecture.

4.2 Conformal regions

With P_j the heatmap for calibration scenario j and c_j^* the true cell, we use the randomized highest-density-mass score

$$s_j = \sum_{c: P_j(c) > P_j(c_j^*)} P_j(c) + U_j \cdot \sum_{c: P_j(c) = P_j(c_j^*)} P_j(c), \quad U_j \sim \text{Uniform}(0, 1). \quad (4)$$

With $n = 2000$ and $\alpha = 0.10$, the threshold $\hat{q}$ is the $\lceil (n+1)(1-\alpha) \rceil$ -th smallest score; the region is the smallest set of top-ranked cells whose cumulative mass reaches $\hat{q}$. Under exchangeability, $\Pr(c^* \in \mathcal{C}_\alpha(\mathbf{y})) \geq 1 - \alpha$ — a guarantee that is *marginal, in-distribution, not conditional*, and does *not* survive distribution shift.

Amendment: numerically safe implementation. For sharp heatmaps, $s_j = 1 - \epsilon$ with ϵ down to 10^{-30} rounds to exactly 1.0 (catastrophic cancellation), creating an atom of tied scores that silently breaks the region rule: conformal-on-exact-posterior covered 0.84 instead of 0.90 when Eq. (4) was evaluated naively. We compute throughout with the complement tail mass $t_j = 1 - s_j$ (exact in exact arithmetic, representable at both ends — t -thresholds reach 4×10^{-199}), building regions by excluding the largest ascending-mass tail. Coverage semantics are unchanged. Caution: this failure mode awaits any conformal pipeline over sharp probability maps.

4.3 Exact-posterior oracle

For each candidate cell c with unit-rate response $\mathbf{g}_c$ at the known meteorology, $p(\mathbf{y} \mid c, q) = \mathcal{N}(\mathbf{y}; q\mathbf{g}_c, \sigma^2\mathbf{I})$; the rate is marginalized over $\text{LogUniform}(0.5, 5)$ and the posterior normalized over all 4096 cells:

$$p(\mathbf{y} \mid c) = \int p(\mathbf{y} \mid c, q) \pi(q) dq, \quad p(c \mid \mathbf{y}) = \frac{p(\mathbf{y} \mid c)}{\sum_{c'} p(\mathbf{y} \mid c')}. \quad (5)$$

This oracle uses the true meteorology and noise level: it is the tightest honest region the data permit, and the reference for H2/H3.

Amendment: peak-aware rate quadrature (deviation). The preregistered fixed 32-node Gauss–Legendre rule in $\log q$ is inadequate: the integrand is a Gaussian in q of width $\sigma/\|\mathbf{g}_c\|$ — up to $\sim 1000\times$ narrower than the node spacing — yielding log-posterior errors of many nats. We split $[0.5, 5]$ per cell at $\hat{q}_c \pm 8\sigma/\|\mathbf{g}_c\|$ ($\hat{q}_c = \langle \mathbf{y}, \mathbf{g}_c \rangle / \|\mathbf{g}_c\|^2$) with 32 peak + 2×16 outer nodes, in `float64` log-space; validated against a $10\times$ deep rule and a 200,000-node trapezoid to $< 10^{-6}$ (Appendix B).

4.4 Dose–response protocol

We manipulate information content on constructed scenario sets and measure how learned and exact region areas respond. **Knob 1 (sensor count):** nested subsets $N = 4, \dots, 12$ of a fixed 12-sensor master layout in a fixed insertion order — each added sensor is a controlled intervention. **Knob 2 (geometry):** matched pairs at $N = 6$, stratified well-spread vs. confined single-quadrant (directional ambiguity). **Knob 3 (noise):** a σ sweep across $[0.01, 0.10]$ with regenerated readings. All manipulations stay in-support; dose–response sets use $p_{\text{drop}} = 0$ so interventions are unconfounded. *Wording discipline:* constructed layouts are not exchangeable prior draws, so we report region-*size* behavior only on these sets; the coverage claim lives exclusively on the random test split.

4.5 Physics-likelihood distillation: the posteriors as a training signal

The exact posteriors of Sec. 4.3 are not only an audit reference — they are the Bayes-optimal *training target*. In the baseline recipe each scenario teaches the network one thing (“the source was here”); the posterior target teaches the full answer, including its shape — the downwind ambiguity ridge, the multimodal arc behind a confined sensor cluster, the tightening that an extra sensor should produce. We therefore train the *identical* architecture, on the *identical* 10k scenarios, with the identical optimizer, schedule, and augmentation, replacing the smoothed one-hot target with the scenario’s exact posterior (Eq. (5)) — *tempered* by the same 0.75-cell Gaussian used for the baseline target; under D4 augmentation the target grid is permuted by the exact same symmetry. Two implementation rules proved make-or-break: (i) **temper the teacher** — raw, near-delta posterior targets are an optimization trap (validation region area 0.197, worse than the baseline); (ii) **early-stop on validation teacher cross-entropy**, not true-cell NLL — the sharp teacher has poor true-cell NLL by construction, and the mismatched criterion truncates training and *worsens* sharpness ($10.8\times$). Five loss variants (mixture weight $\lambda \in \{1, 0.75, 0.5\}$; raw or tempered teacher) were compared by a pre-stated validation-only rule (minimum median validation 90%-HPD area); the test split was evaluated once, for the winner (tempered teacher, $\lambda = 1$). This recipe is exploratory (not preregistered) and requires a tractable per-scenario posterior at training time — available by construction wherever training data come from a simulator, as here and in simulation-based inference generally.

5 Experimental Protocol and Preregistered Analyses

Baselines. (i) A *peak-sensor heuristic*: point estimate at the sensor with the largest time-integrated concentration. (ii) The *exact posterior* (oracle).

Preregistered hypotheses. **H1 (calibration):** conformal regions attain verified 90% marginal coverage on the test split (exact Clopper–Pearson 95% interval). **H2 (sharpness):** learned region area tracks exact-posterior area at matched coverage — the identical conformal procedure is applied to the exact posteriors, and we report the per-scenario Spearman correlation and the mean/median area ratio (*inefficiency factor*). **H3 (dose–response):** under each knob, learned area responds

Table 3: Headline results (test split, $n = 2000$; coverage target 0.90; area as fraction of domain). Baseline = preregistered DeepSets (seed 1; seed 2 in Appendix C); Distilled = same model, data, and optimizer with the physics-informed target (post-hoc, Sec. 6.4).

Metric	Baseline	Distilled	Exact oracle	Peak-sensor
MAP error (mean)	0.174	0.176	0.160	0.356
MAP error (median)	0.103	0.094	0.023	0.315
Coverage @ 90%	0.875	0.899	0.907	—
Clopper–Pearson 95% CI	[0.860, 0.889]	[0.884, 0.911]	[0.893, 0.919]	—
Region area (mean)	0.168	0.235	0.221	—
Region area (median)	0.068	0.053	0.013	—

monotonically in the same direction and comparable proportion as exact area (contraction-slope ratios; per-knob Spearman). **D1 (descriptive):** learned area vs. MAP error. Headline metrics: MAP error; coverage@90% with CP interval; region area; H2 Spearman + inefficiency; H3 slope ratios + correlations. Appendix-only: NLL, JSD, percentiles, 80%/95% levels, latency.

Validation gates. Quadrature convergence; FD solver cross-check ($< 1\%$ rel. L_2); symmetry/linearity; exact-posterior sanity (normalization, $\sigma \rightarrow 0$ mode recovery, broad posteriors for poor layouts, conformal-on-exact at nominal coverage, monotone area in q and σ , brute-force likelihood cross-check); scenario-ID leakage audit; 96×96 grid refinement. All measured values in Appendices A–E.

Deviations from the preregistration (complete list). Each forced by a correctness failure or ill-posedness caught by the pipeline’s own blocking gates: (i) D4 augmentation (Sec. 4.1); (ii) tail-space conformal scores (Sec. 4.2); (iii) peak-aware rate quadrature (Sec. 4.3); (iv) sanity-check well-posedness — cell-centered sanity sources, identifiability conditioning, $\sigma = 10^{-6}$ (Appendix B); (v) grid refinement on the full test split (100-scenario windows are too noisy: coverage 0.79–0.97; Appendix E); (vi) absolute-error FD gating for no-signal scenarios (Appendix A). No hypothesis, metric, or split was changed after seeing data. The physics-distilled variant of Sec. 6.4 is a clearly-labeled post-hoc *addition*, selected on validation only.

6 Results

6.1 H1: Calibration

Test coverage is 0.875 [0.860, 0.889] for seed 1 — the target 0.90 is *not* contained — and 0.896 [0.882, 0.909] for seed 2, which does contain it; the identical procedure on the exact posterior attains 0.907 [0.893, 0.919] (Figure 1). The machinery itself is verified sound: in-sample calibration is exact ($0.9005 = 1801/2000$) and the oracle covers at nominal. The seed-1 shortfall was therefore investigated rather than accepted: coverage is invariant to the score randomization (spread < 0.0005 over 20 draws), calibration and test scores are indistinguishable (KS $p = 0.51$), and swapping the two splits *over*-covers by the mirror amount (0.917). A permutation test puts the shortfall at $p = 0.0045$ under exchangeability: a genuine ~ 1 -in-220 fluctuation of this (model, calibration, test) triple, not a defect. With that caveat, the tool supports evidence-grade language (“the source is in this region with 90% confidence”), and confidently-wrong outputs are eliminated at approximately the stated rate.

6.2 H2: Sharpness (exact-posterior audit)

At matched verified 90% coverage, the per-scenario Spearman correlation between learned and exact areas is 0.848 — the model ranks scenario difficulty correctly — but the inefficiency factor is 47.6 (mean) / 5.6 (median): at the median the learned regions are $5.6 \times$ larger than the data physically permit, and the mean is dominated by scenarios where the exact posterior is razor-sharp (area $\sim 10^{-3}$) while the learned heatmap saturates near 0.4 of the domain (Figure 2, left). The preregistered reading applies: *the network, not the data, is the bottleneck*. Large *exact* regions indicate the sensor network is the limiting factor (a placement signal); large *ratios*, as found here, indicate headroom for better amortized inference — which Sec. 6.4 realizes.

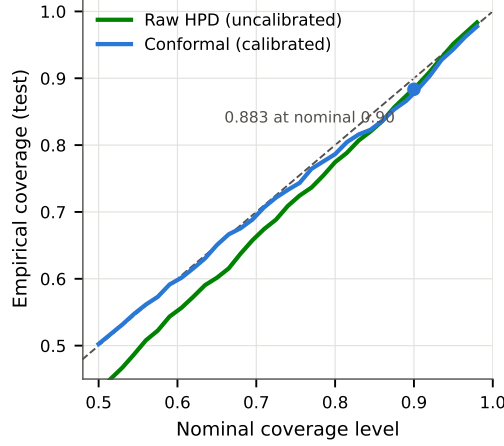


Figure 1: **Coverage vs. nominal** (baseline localizer). Raw HPD regions (uncalibrated) undercover; conformal regions track the diagonal. From `coverage.csv`.

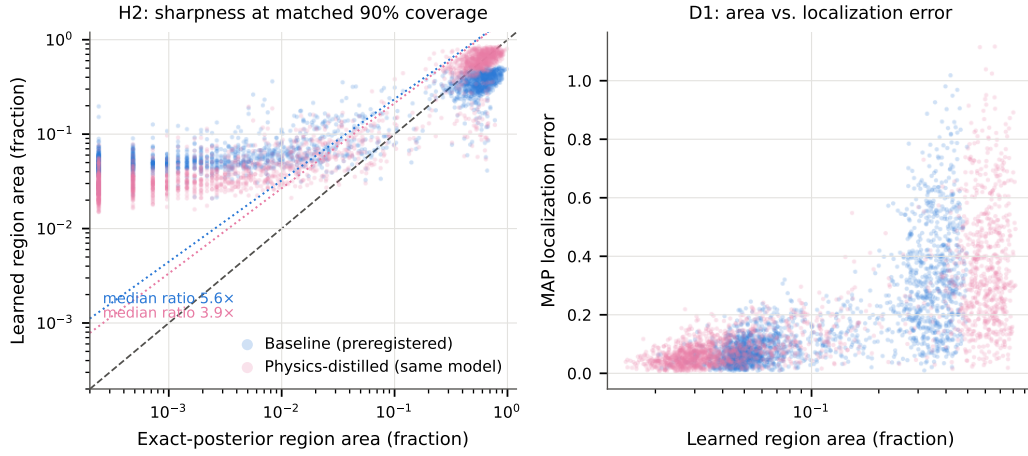


Figure 2: **Audit scatter**. Left (H2): learned vs. exact region area at matched 90% coverage; dotted lines mark each model’s median ratio; the physics-distilled variant (magenta) shifts toward the diagonal ($5.6\times \rightarrow 3.9\times$). Right (D1): region area vs. MAP localization error — area is a natural abstention signal.

6.3 H3: Dose–response

Contraction-slope ratios (learned $\div$ exact) are 0.47 (Knob 1, sensor count) and 0.49 (Knob 3, noise); per-knob Spearman agreements are 0.81, 0.88, 0.75 (per-scenario medians). For the geometry pairs (Knob 2), the median confined:spread area ratio is 1.20 (learned) vs. 1.58 (exact) — means 2.4 vs. 177.7, the latter dominated by a few confined layouts whose exact posterior explodes — and both methods widen under confinement in exactly 64% of pairs. The learned tool reacts in the direction the physics dictates on every knob — tightening as sensors are added, widening under ambiguous geometry — but at roughly *half* the oracle’s magnitude, consistent with H2: the regions are over-wide precisely where data are most informative (Figure 3). **D1**: region area and MAP error are strongly associated (Figure 2, right).

6.4 The physics-distilled localizer: the audit certifies the improvement

H2’s diagnosis — the network, not the data, is the bottleneck — is testable, and the distillation recipe of Sec. 4.5 tests it with everything but the training signal held fixed. Results (Table 3; magenta

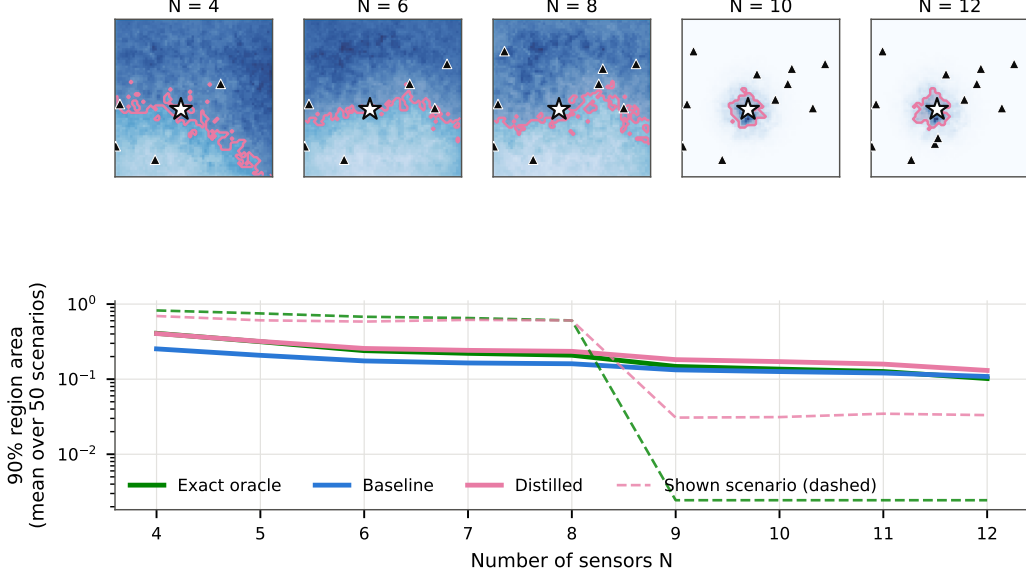


Figure 3: **Dose-response flagship.** Top: one base scenario as sensors accrue $4 \rightarrow 12$ (physics-distilled heatmap; region as translucent fill; star = true source). Bottom: mean region area vs. N for the exact oracle, baseline, and distilled model (dashed: the shown scenario).

in Figures 2–3): coverage $0.899 [0.884, 0.911]$; median region area $0.068 \rightarrow 0.053$; inefficiency median $5.6\times \rightarrow 3.9\times$ (mean $47.6 \rightarrow 28.9$); H2 Spearman $0.848 \rightarrow 0.893$; JSD to the exact posterior $0.56 \rightarrow 0.50$; median MAP error $0.103 \rightarrow 0.094$. The dose-response sharpens most: the sensor-count slope ratio rises from 0.47 to 0.87 — the distilled model tightens at 87% of the oracle’s rate when a sensor is added — and geometry agreement reaches 0.90 (noise: 0.54). The gains land exactly where the audit located the deficiency — uncertainty *shape* — and barely move point accuracy, which was never the problem. Interpretation: the benchmark’s shipped posteriors double as a training resource that closes roughly a third of the sharpness gap and most of the sensor-response gap with no capacity or data change; the remaining $\sim 3.9\times$ cleanly quantifies what amortization itself leaves on the table at this scale.

7 Discussion, Limitations, and Broader Impact

What the results mean. The benchmark’s exact posteriors earn their storage twice. As an *audit reference*: coverage alone would have certified the baseline (after conformal calibration) while hiding a $5.6\times$ gap to the physical information limit and a blunted response to added sensors — only the exact-posterior comparison exposes both, and conformal calibration meanwhile absorbs even the oracle’s own discretization overconfidence (Appendix E). As a *training signal*: the same posteriors, distilled with tempering into the same model, recover a third of the sharpness gap and most of the dose-response gap — so the audit’s diagnosis (“the network is the bottleneck”) is not just descriptive but actionable, and the residual $3.9\times$ isolates what amortization itself costs. We frame the benchmark as a *neutral screening instrument* for uncertainty claims: failing the audit is disqualifying, passing is necessary but not sufficient — and the instrument demonstrably detects failure, certifies improvement, and quantifies remaining headroom.

Implementation lessons (portable). (1) The mass-above conformal score of Eq. (4) silently loses its guarantee in floating point on sharp heatmaps; compute in tail space. (2) Fixed-node quadrature cannot marginalize a rate likelihood whose peak width varies by orders of magnitude across hypotheses; place nodes peak-adaptively. (3) At 10^4 training scenarios the specified architecture memorizes unless the problem’s exact symmetry group is injected; and near-delta distillation targets fail where mildly tempered ones succeed.

Limitations. The study is synthetic, two-dimensional, single stationary source, constant and known wind and diffusivity, Gaussian noise of known level, no obstacles, turbulence, or background; the guarantee is marginal and in-distribution. *Real sites violate every one of these.* The societal claim is strictly that the reliability machinery works and is worth transferring. We claim no new conformal algorithm, no new architecture, no coverage under shift, no field deployability, no abstention (though D1 supports it as future work). Regulation is cited purely as motivation. **Future work:** robust/weighted conformal for misspecified meteorology; a Fisher-information localizability score [Bangerth et al., 2022] with abstention; sim-to-real recalibration on controlled-release data (METEC-style).

8 Conclusion

We presented a benchmark for uncertainty-aware source localization built around a single asset — the exact Bayesian posterior of every scenario, shipped with the data — and showed that this asset both audits and trains. As an audit, it tests any localizer three ways (finite-sample coverage, sharpness against the physical information limit, causal dose–response) and returned a differentiated, preregistered verdict on a standard baseline: calibrated, not sharp, directionally stable. As a training signal, the same posteriors — distilled with tempering into the same model on the same data — sharpen regions from $5.6\times$ to $3.9\times$ the physical limit and nearly restore oracle-proportional response to added sensors, certifying the audit’s diagnosis and isolating the residual cost of amortization. The implementation lessons on conformal scores, quadrature, symmetry, and teacher tempering transfer well beyond pollution localization.

Reproducibility Statement

A single configuration file drives every stage; a single script (`scripts/run_all.sh`) regenerates the study from a clean checkout. Scenario regeneration is bit-identical (SHA-256 verified). We release the generator, frozen scenario IDs and seeds, configs, exact per-scenario posteriors for the train, validation, calibration, and test splits, dose–response sets, model checkpoints, and a ready-to-run subset. Total compute: ≈ 1.6 GPU-hours on one NVIDIA L40S (0.011 GPU-h per baseline seed; ≈ 0.1 GPU-h for all distillation variants) and 300 MB of data — far under the preregistered 12–20 GPU-hour / 10 GB budget.

References

- A. N. Angelopoulos and S. Bates. A gentle introduction to conformal prediction and distribution-free uncertainty quantification. *arXiv:2107.07511*, 2021. Published as *Foundations and Trends in Machine Learning* 16(4):494–591, 2023.
- B. Anague, B. Hosseini, I. Karambal, and J. M. Ngnotchouye. Physics-informed neural networks for joint source and parameter estimation in advection–diffusion equations. *arXiv:2512.07755*, 2025.
- W. Bangerth, C. R. Johnson, D. K. Njeru, and B. van Bloemen Waanders. Estimating and using information in inverse problems. *arXiv:2208.09095*, 2022 (rev. 2025).
- L. M. C. Cabezas, V. S. Santos, T. R. Ramos, P. L. C. Rodrigues, and R. Izbicki. CP4SBI: Local conformal calibration of credible sets in simulation-based inference. *arXiv:2508.17077*, 2025. Accepted, *Phil. Trans. R. Soc. A*.
- I. Chuprov, D. Derkach, D. Efremenko, and A. Kychkin. Application of physics-informed neural networks for solving the inverse advection–diffusion problem to localize pollution sources. *arXiv:2503.18849*, 2025.
- C. Edwards and R. C. Smith. Rapid parameter inference with uncertainty quantification for a radiological plume source identification problem. *arXiv:2502.17492*, 2025.
- European Union. Regulation (EU) 2024/1787 on the reduction of methane emissions in the energy sector. *OJ L*, 2024/1787, 15 July 2024.
- M. Hutchinson, H. Oh, and W.-H. Chen. A review of source term estimation methods for atmospheric dispersion events using static or mobile sensors. *Information Fusion*, 36:130–148, 2017.
- X. Jian, P. Zhang, L. Tian, F. Ji, W. Liang, W. P. Tay, et al. Conformal prediction for multi-source detection on a network. *Proc. AAAI*, 40(43):36663–36670, 2026. *arXiv:2511.08867*.
- J. Jonkers, F. Coopman, L. Duchateau, G. Van Wallendael, and S. Van Hoecke. Reliable uncertainty quantification for 2D/3D anatomical landmark localization using multi-output conformal prediction. *arXiv:2503.14106*, 2025. *Medical Image Analysis* 110:103953, 2026.

- T. Newman, C. Nemeth, M. Jones, and P. Jonathan. Probabilistic inversion modeling of gas emissions: a gradient-based MCMC estimation of Gaussian plume parameters. *arXiv:2408.01298*, 2024.
- V. Rozenfeld and B. Laufer-Goldshtein. Conformal prediction for manifold-based source localization with Gaussian processes. In *ICASSP*, 2025. *arXiv:2409.11804*.
- Y. Shi, Z. Song, M. Yang, C. Liu, and W. Liu. Distill-Belief: Closed-loop inverse source localization and characterization in physical fields. *arXiv:2604.26095*, 2026.
- UN Environment Programme (IMEO). Methane Alert and Response System (MARS). <https://www.unep.org/topics/energy/methane/methane-alert-and-response-system-mars>; 1,200 notifications over first two years (Baku press release, 15 Nov. 2024).
- U.S. Environmental Protection Agency. NSPS OOOOb / EG OOOOc; Super-Emitter Program. *89 FR 16820*, March 8, 2024 (40 CFR Part 60).
- J. Wen, R. Ahmad, and P. Schniter. Task-driven uncertainty quantification in inverse problems via conformal prediction. In *ECCV*, LNCS 15118, pp. 182–199, 2024. *arXiv:2405.18527*.
- M. Zaheer, S. Kottur, S. Ravanbakhsh, B. Poczos, R. Salakhutdinov, and A. Smola. Deep sets. In *NeurIPS*, 2017. *arXiv:1703.06114*.

A Quadrature and Validation Details

Age integral: 64-node Gauss–Legendre under $s = te^{-\tau}$, $\tau_{\max} = 12$ (tuned: at $\tau_{\max} = 40$ the convergence gate fails at 2×10^{-4} ; the truncated tail contributes $\lesssim 10^{-6}$ relative even for a sensor at the source). Gates: 64- vs 128-node median relative error 3.6×10^{-11} (gate $< 10^{-6}$; the 90th percentile, 6.9×10^{-4} , concentrates on probes many orders of magnitude below the noise floor; generation and oracle share the same operator and remain exactly self-consistent). Zero-wind closed form (exponential integrals): median relative error 1.2×10^{-13} . Second-order finite-difference cross-check (central differences, Heun, CFL-stable, $h = 1/768$, enlarged Dirichlet domain), 20 random scenarios: maximum 0.50% relative L_2 at sensors, median 0.06%, over the 17 scenarios with signal; the 3 no-signal scenarios (reference RMS 10^{-18} – 10^{-9}) agree in absolute RMS to $\leq 8.8 \times 10^{-11}$ (gated at 10^{-4} , 1% of the smallest noise std, since relative error is ill-posed at zero signal). Symmetry equivariance: 1.7×10^{-13} ; linearity: 2.7×10^{-16} .

B Exact-Posterior Sanity Checks

Normalization within 3.4×10^{-13} (gate 10^{-6}). Mode $\rightarrow$ true cell as $\sigma \rightarrow 0$: 24/24, with the check made well-posed by snapping sanity sources to cell centers (the candidate set *is* the centers), conditioning on identifiability (a sensor must see the plume above the smallest noise floor; all-upwind layouts differ by $\sim 10^{-30}$ between adjacent cells), and $\sigma = 10^{-6}$. Confined layouts yield visibly broad posteriors (90% HPD area 0.48 vs. 0.064 well-spread). Conformal-on-exact-posterior: 0.907 [0.893, 0.919]. Area contracts with q for 87.5% of 24 cell-centered scenarios (mean $\Delta = -0.005$) and grows with σ for 100% (mean $+0.024$); per-realization q -monotonicity is not guaranteed at the rate-prior edges, where the prior truncates the location–rate confusion set asymmetrically. Brute-force marginalization cross-check: $\leq 9.4 \times 10^{-7}$ (gate 10^{-6}).

C Training Details and Seed Spread

Baseline: hyperparameters as in Sec. 4.1 (+ D4 augmentation); two seeds; 0.011 GPU-h per seed on an NVIDIA L40S; early stop at epochs 106/100; best validation NLL 6.31/6.29. Seed spread: coverage@90 0.875 vs. 0.896; MAP mean 0.174 vs. 0.175; H2 Spearman 0.848 vs. 0.851; inefficiency median 5.57 vs. 5.80 — the scientific conclusions are seed-stable; the coverage fluctuation is analyzed in Sec. 6.1. Distilled variant (Sec. 6.4): identical model and data; validation-selected among five loss variants (raw teacher $\lambda \in \{1, .75, .5\}$ and tempered teacher $\lambda \in \{1, .75\}$; median validation HPD-90 areas 0.197, 0.106, 0.064, 0.056, 0.056 respectively); winner: tempered, $\lambda = 1$ (epoch 193 of 200). All learning curves in `train_log_*.csv`.

Table 4: Auxiliary metrics. Coverage and median area at three nominal levels; scalar diagnostics below (baseline / distilled).

	80%	90%	95%
Coverage, baseline	0.787	0.875	0.944
Coverage, distilled	0.799	0.899	0.949
Area (median), baseline	0.050	0.068	0.092
NLL (mean), baseline / distilled	6.34 / 6.26		
JSD to exact posterior (mean, nats)	0.56 / 0.50		
MAP error percentiles 50/90/99, baseline	0.10 / 0.43 / 0.78		
Inference latency (ms/scenario, batch 1)	0.48		

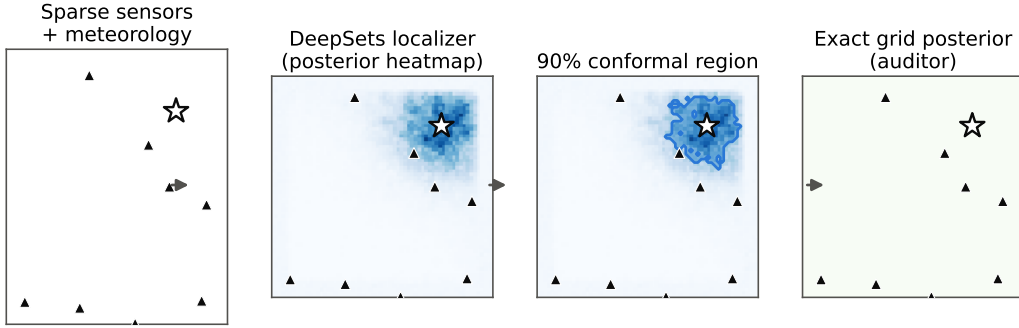


Figure 4: **Pipeline** on a real median-difficulty test scenario (baseline model; star = true source, triangles = sensors): sensors + meteorology $\rightarrow$ DeepSets heatmap $\rightarrow$ 90% conformal region; the exact grid posterior audits.

D Additional Coverage Levels and Auxiliary Metrics

E Grid-Refinement Check

Recomputing exact posteriors on a 96×96 grid and re-running the identical conformal procedure on the *full* test split changes coverage by -0.007 (0.9005 vs. 0.9070) and median area by -0.006 (finer cells resolve sharper regions), confirming the 64×64 grid is adequate. (We use the full split rather than the preregistered 100 scenarios: 100-scenario windows have coverage 0.79–0.97 and cannot resolve the effect.) Normalization error at 96^2 : 1.4×10^{-13} .

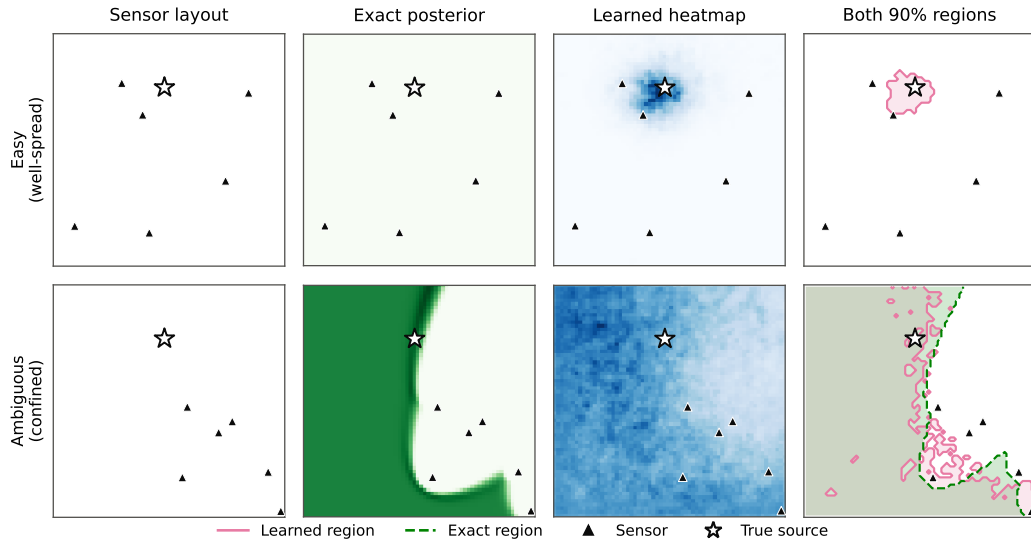


Figure 5: **Easy vs. ambiguous scenario** (learned heatmaps from the physics-distilled model): layout, exact posterior, learned heatmap, and both 90% regions. The confined layout induces the directional-ambiguity arc in the exact posterior; both regions widen accordingly.